

Selective spatial atomic layer deposition of Cu, Cu₂O, and CuO thin films in the open air: reality or fiction?

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ABSTRACT

Copper and binary copper oxide thin films are key materials for microelectronic, optoelectronic, and sensing devices. Herein, we report innovative atmospheric pressure spatial atomic layer deposition processes enabling the facile control over the copper oxidation state. The selective deposition of Cu, Cu₂O, and CuO thin films at low temperatures (160–260 °C) has been achieved by using Cu(1)(hfac)(tmvs) as copper source and nitrogen, water, or ozone as coreactants, respectively. The three obtained materials are pure and crystalline, as revealed by X-ray absorption spectroscopy, X-ray photoelectron spectroscopy, and X-ray diffraction. The deposition mechanism of the Cu, Cu₂O, and CuO phases is based on disproportionation, hydrolysis, and oxidation reactions. Interestingly, metallic copper films with resistivity as low as $\sim 3.7 \times 10^{-6} \Omega \cdot \text{cm}$ have been obtained, and resistivity values of $100 \Omega \cdot \text{cm}$ and $0.2 \Omega \cdot \text{cm}$ have been measured for the Cu₂O and CuO semiconductor layers, respectively. Although working in an open-air environment, the oxidation state of copper-based thin films can be controlled using the same precursor, thus reducing cross-contamination and allowing for a faster and more reliable process. This demonstrates the versatility of atmospheric-pressure spatial atomic layer deposition and opens prospects for the manufacturing of novel devices such as wearables, sensors, or flexible electronics.

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1. Introduction

The performance of optoelectronic and microelectronic devices depends mainly on the properties of active functional materials such as metals, semiconductors, and insulators [1–4]. Thanks to its low resistivity and high thermal stability against electromigration [5], copper has been widely used as an interconnecting and electrode material for several applications [6–9], while semiconducting copper oxides are appealing for photovoltaics, catalysis, and sensors [10–15] thanks to their electro-optical properties and the relative abundance of copper.

Metallic copper has been deposited using different approaches in the past, and most of the industrial processes used for copper metallization are based on chemical vapor deposition (CVD) [16], electroless deposition [17], solution based [18], and electroplating [19,20]. Other chemical approaches based on the atomic layer deposition (ALD) method have been reported in the literature [21–23]. As for metallic copper, copper oxide thin films (with tunable properties) have been obtained by different approaches, such as sputtering, PLD, thermal annealing, and electrodeposition [24], providing different properties and oxidation states of the material [11]. Similarly, ALD has been used recently to deposit Cu₂O thin films for several applications [25–30]. Alternatively, Cu₂O could also be obtained by performing a postannealing treatment at different temperatures (≤ 200 °C) to promote the oxidation of the metallic copper [31,32]. This temperature is further increased (300–500 °C) in order to enhance the oxidation kinetics that allows the achievement of the CuO phase [32,33].

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Among the different thin film deposition techniques, ALD is unique since it allows extreme film thickness control below the nanometer and provides conformal coatings over high-aspect-ratio features [34]. This is due to surface-limited, self-terminated reactions taking place through the sequential exposition of the surface to different precursors, with intermediate purging steps [34–36]. Nowadays, ALD allows the synthesis of a variety of materials, such as oxides [37,38], nitrides [39,40], and metals [41], and the technique has gained momentum in the last years for the deposition of metallic copper [34,36].

The ALD of metallic copper has been achieved using different chemistries. For instance, Knisley et al. have studied the formation of pure copper films at 120 °C using a three-step ALD process including $\text{Cu}(\text{OCHMeCH}_2\text{NMe}_2)_2$, formic acid, and hydrazine [42]. Conversely, Kalutarage et al. proposed a two-step process in the temperature window 100–180 °C using $\text{Cu}(\text{OCHMeCH}_2\text{NMe}_2)_2$ and borane dimethylamine as a reducing coreactant on Ru substrates [43]. Alternatively, other studies have been performed on the deposition of copper films using different precursors and hydrogen gas as coreactant in a single-step method [44–46]. Besides thermal activation, different plasma-assisted processes have also been investigated to allow a lower growth temperature of copper (≤ 130 °C) based on the ALD approach [47–49].

The deposition of binary copper oxides by ALD has already been evaluated. Several processes have been reported for the deposition of Cu_2O , through the hydrolysis of precursors containing Cu^+ ions [28], or the hydrolysis and thermally induced reduction of precursors containing Cu^{2+} ions [30,50] (similarly to what is observed for MOCVD processes where Cu_2O is obtained from precursors containing Cu^{2+} ions without the need of reducing atmospheres) [51]. CuO thin films have been obtained using ozone or oxygen plasma as coreactants [14].

Despite the unique assets of ALD, it is a rather slow technique that is usually performed in vacuum, thus limiting its industrial appeal. Atmospheric-pressure spatial ALD (AP-SALD) is a promising candidate to achieve a fast and scalable deposition of functional thin films, providing similar quality as conventional ALD [52–54]. Spatial ALD is an approach where the precursors are not injected temporally but rather are constantly being supplied and kept apart (i.e., spatial separation) by inert gas barriers, while the sample is exposed to the different precursor zones. Metallic copper has recently been obtained for the first time using this technique with a carbene-stabilized Cu precursor and hydrogen plasma in a system within a controlled inert atmosphere [55].

Thus, while different approaches have been proposed for the deposition of Cu, Cu_2O , and CuO thin films, no reports exist on the deposition of the three phases using the same approach, and different postannealing treatments are used to convert the pristine films to the other phases. For example, Lenef et al. deposited Cu_2O films using hydrogen and oxygen plasma, while Cu and CuO were obtained by inducing a postannealing treatment of 600 °C in vacuum (for Cu) and 350 °C in the air (for CuO) [49]. Likewise, Jürgensen et al. investigated the deposition of metallic copper using the MOCVD technique at 400 °C on several substrates and then performed a postannealing treatment at 500 °C for 2 h to promote the formation of the CuO phase. Moreover, they also used a PECVD technique with either oxygen or hydrogen to respectively grow Cu and Cu_2O thin films that were then converted to CuO films with a postannealing treatment [56]. Thus, the direct selective deposition of the three different phases without using a reducing agent or plasma and an annealing step is a challenging task, in particular when low-temperature processing is required for a particular application.

In this work, we show for the first time, to the best of our knowledge, that AP-SALD can be used as a low-temperature and open-air approach to selectively deposit Cu, Cu_2O , or CuO in a single

step using the same Cu precursor and only changing the coreactant (H_2O for Cu_2O , O_3 for CuO , and N_2 for Cu). To the best of our knowledge, this is the first time that CuO thin film has been deposited using an AP-SALD system. Remarkably, metallic Cu films are obtained at low temperatures, without the need for a sealed vacuum chamber and the addition of a reducing atmosphere or plasma. Since the deposition of Cu_2O has been already reported in the literature [57–59], here, we will focus mainly on the CuO and Cu phases [57].

2. Experimental

2.1. Deposition of metallic copper, CuO , and Cu_2O

The deposition of the three phases was performed using an open-air and scalable SALD approach based on a manifold head, as shown in [Supplementary Fig. S1](#). The latter contains one metal precursor outlet, separated from the coreactant outlets by inert gas barriers alternatively located in between the reactive gas flows to prevent them from mixing [60]. The precursor, $\text{Cu}(\text{I})(\text{hfac})(\text{tmvs})$ (Epivalence, Limited) was heated at 65 °C and the transport line between the bubbler and the manifold injection head was heated at 70 °C to avoid any condensation of the precursors in the lines. A mass flow of 60 sccm was used for the Cu precursor, which was diluted in extra nitrogen flow (60 sccm) before being sent to the injection head. A thermogravimetric analysis was performed to assess the decomposition and the volatility of the precursor (shown in [Fig. S2](#)). The thermogravimetric analysis measurement revealed a gradual evaporation starting from 50 °C, which implies that the precursor has a sufficient vapor pressure at 60–70 °C to permit ALD growth, as shown from previous studies [28,57]. Further, water was carried using a mass flow of 120 sccm, carried in an additional 120 sccm N_2 flow from two lines, each delivering 120 sccm to the glass substrate. The ozone was generated by a home-made ozone generator powered by a pulsed high voltage, in which the oxygen flow rate and the high voltage pulse frequency were varied to adjust the amount of ozone reaching the deposition area (it will be referred to as “ozone rate”, OR). The voltage was fixed at 5.5 kV, and then frequency was adjusted to obtain the desired OR. Finally, the nitrogen barrier flow was adjusted to obtain a flow of 120 sccm per outlet. The precursor and coreactants were injected continuously from the manifold head onto the substrate that oscillated at a scanning speed of 8 cm/s for Cu and 10 cm/s for CuO and Cu_2O . The displacement for Cu was limited to 4 mm preventing the exposure of the growing film to the atmosphere, while the displacement for CuO and Cu_2O was 70 mm. Films were deposited on borosilicate glass substrates with a surface area of $3 \times 5 \text{ cm}^2$. The substrates were cleaned with isopropanol in an ultrasonic bath for 5 min and then dried with N_2 . The gap between the head and the substrates was maintained at a distance of 150 μm in all cases.

2.2. Characterization

Film thickness was estimated by ellipsometry (using a compact Film Sense FS-1 ellipsometer). The surface morphology and thickness of the thin films were evaluated by scanning electron microscopy (SEM-FEG, GeminiSEM 300). Atomic force microscopy was used on the different layers in tapping mode using a Digital instrument D3100 Nanoscope. Transmission electron microscopy (TEM) imaging was performed with a JEOL JEM 2010 microscope. Cross-sections were prepared by automated polishing, the latter using the MultiPrepTM system (Allied High Tech Products, Inc.). X-ray diffraction (XRD) patterns were obtained with a Bruker D8 Advance diffractometer in the Bragg-Brentano (θ – 2θ) configuration, with Cu $\text{K}\alpha 1$ radiation (0.15406 nm). Raman measurements

were obtained in the wavenumber range of 50–700 cm^{-1} using a Jobin Yvon/HoribaLabRam spectrometer equipped with a blue laser (488 nm). The optical properties of the films were evaluated with a lambda 950 UV–Vis spectrophotometer from Perkin Elmer within the range 250–2500 nm. The optical bandgaps of the different films were extracted from the transmittance and reflectance spectra using the Tauc plot method [61,62]. X-ray photoelectron spectroscopy (XPS) measurements were performed using a Thermo-Scientific K-Alpha spectrometer with an Al K α (1486.6 eV) X-ray source. The samples were subjected to an Ar milling process to remove surface contaminants at 2 keV for 10 and 40 s. The electrical properties were measured with a Keithley 2400 source meter connected to a 4-point probe station in the Van der Pauw configuration at room temperature. X-ray absorption near edge structure and extended X-ray absorption fine structure were investigated at BM25-SpLine beamline at the ESRF at the Cu K-edge (8978.9 eV) in fluorescence mode ranged from 8900 to 9200 eV. X-ray absorption spectroscopy (XAS) signatures were acquired using a HITACHI Vortex®-EM Silicon Drift Detector (SDD), and XAS spectra were analyzed considering the standard procedures using the ATHENA software package [63].

3. Results and discussion

As indicated above, Cu(I) (hfac) (tmvs) has been studied in the past for the deposition of Cu interconnects [64,65]. The bonding between the different chemical species of the precursor has been revealed through numerical simulations [66]. An interesting investigation has explored the activation energy and reaction rate of the precursor to produce copper and consequently reported that delamination problems at the interface remain while depositing metallic copper at 200 °C due to the presence of fluorine during the deposition [67].

Despite the technical problems encountered with this specific precursor to deposit metallic copper, the mechanism is well established in the literature. Indeed, the relative low stability of Cu(I)(hfac)(tmvs) results in a disproportionation reaction taking place when subjected to high temperatures (above 60 °C) [68]. In the reaction, two molecules of the precursor react to give one atom of metallic copper and volatile by-products in the form of 2 (tmvs) and Cu(II)(hfac)₂, as shown in the following equation [67]:



As for the two copper oxide phases, Cu₂O has been obtained by hydrolyzing Cu(I) (hfac) (tmvs) with water, as detailed in D. Muñoz-Rojas *et al.* [57]. Finally, the deposition of CuO thin films is expected through the oxidation Cu(I)(hfac)(tmvs) with ozone although it has never been reported.

But it remains to be answered whether the hypothesis of obtaining different copper phases, namely Cu₂O, metallic Cu, and CuO, by changing only the coreactant is possible, in particular using an open-air SALD system. An illustration of the expected three reactions is shown in Fig. 1a, Fig. 1b and 1c and illustrates the responsible mechanisms to obtain the three different phases of the material.

In the following sections, we show that it is indeed possible to selectively deposit Cu/Cu₂O/CuO in the open air from a same precursor and at low temperatures (<260 °C) by just changing the co-reactant used.

3.1. Deposition of metallic copper

Metallic copper films have been deposited on glass substrates by injecting the precursor through the manifold head that is

surrounded by nitrogen flow. This prevents the atmosphere and humidity from reaching the film being deposited on the substrate. The deposition temperature was varied from 160 °C to 240 °C in order to evaluate the effects on the properties of the material. No deposition was observed below 160 °C. Fig. 2a shows SEM and optical pictures of the films. The color of the metallic copper films obtained (see inset in Fig. 2a) evolved from a blue/black color for a deposition temperature of 240 °C, through a black color for 200 °C, and eventually into a reflecting brown copper color at 160 °C. Resistivity was observed to be similar to that of bulk copper ($\sim 4.10^{-6} \Omega \cdot \text{cm}$) for the films deposited at 160 °C and increases until reaching a value of ($\sim 3.10^{-5} \Omega \cdot \text{cm}$) with the rise of deposition temperature (Fig. 2b). The film thicknesses observed (in all cases performing 8000 ALD cycles) were 1 μm , 1.3 μm , and 1.7 μm for samples deposited at 160 °C, 200 °C, and 240 °C, respectively, as measured by cross-section SEM (shown in Supplementary Fig. S3). SEM studies reveal a clear variation in the morphology from one sample to another. Even though copper is stable in air at room temperature and, as reported by Peng *et al.*, the surface modification of the metallic film does not impact the electronic and thermal properties due to the passivation outcome at the surface [69], the oxidation occurs quite easily though by inducing a higher thermal budget [33,56]. We thus expected the changes in morphology, color, and conductivity to be the result of partial oxidation of the deposited film upon increasing the deposition temperature. This was confirmed by further characterization, as detailed below.

The film deposited at low temperature displays a shiny metallic aspect with grains having a diameter of $\sim 0.7 \mu\text{m}$ – $\sim 1 \mu\text{m}$. After increasing the deposition temperature, the film deposited at 200 °C presents agglomeration of small grains, resulting in a rough surface with less dense morphology. At even higher temperatures (240 °C), the morphology changes drastically with respect to sample deposited at 160 °C, with flower-like agglomerates made of small grains. The films were then further analyzed with XRD and Raman spectroscopy measurements. XRD confirmed the crystallinity of the prepared copper films. The three XRD patterns, shown in Fig. 2c, display mainly the metallic copper peaks mostly oriented in the (111) direction and then followed by the (200) direction. The presence of the (111) reflection from Cu₂O is clearly observed for the samples deposited at 200 and 240 °C. On the other hand, Raman spectroscopy, in Fig. 2d, also shows the contribution of the acoustic and optical modes of the Cu₂O phases at 200 and 240 °C. Conversely, and in agreement with XRD results, the sample deposited at 160 °C shows no contribution from Cu₂O in the Raman spectrum.

XPS was performed to define the composition and the oxidation states of films deposited at 160 °C and 240 °C. The measurements were performed without and with Ar milling for 40 s to investigate the presence of the different species at the surface and bulk level. XPS data confirmed the deposition of relatively pure metallic copper films, as no satellites from other phase contributions could be visible in the Cu 2p scan (Fig. 3a). The analysis of the Cu Auger peaks (LMM), shown in Fig. 3b, shows the presence of the Cu¹⁺ for the non-sputtered and sputtered samples. In particular, samples deposited at a lower temperature (160 °C) reveal the evolution of the different species before and after the Ar milling process as well. These data reveal the presence of the Cu¹⁺ in the as-deposited films (without milling), assumed to come from oxidized copper in the surface of the films [70,71]. After Ar milling, the Cu⁰ species becomes prominent for the film deposited at 160 °C. Conversely, the sample deposited at 240 °C still shows the main contribution from Cu¹⁺ even after milling, in agreement with the mixed phase demonstrated in Raman and XRD results. (A complete XPS survey is found in Supplementary Fig. S4).

Fig. 3c and e present the decomposition of the Cu 2p peaks for the samples deposited at 160 °C and 240 °C without Ar milling. The

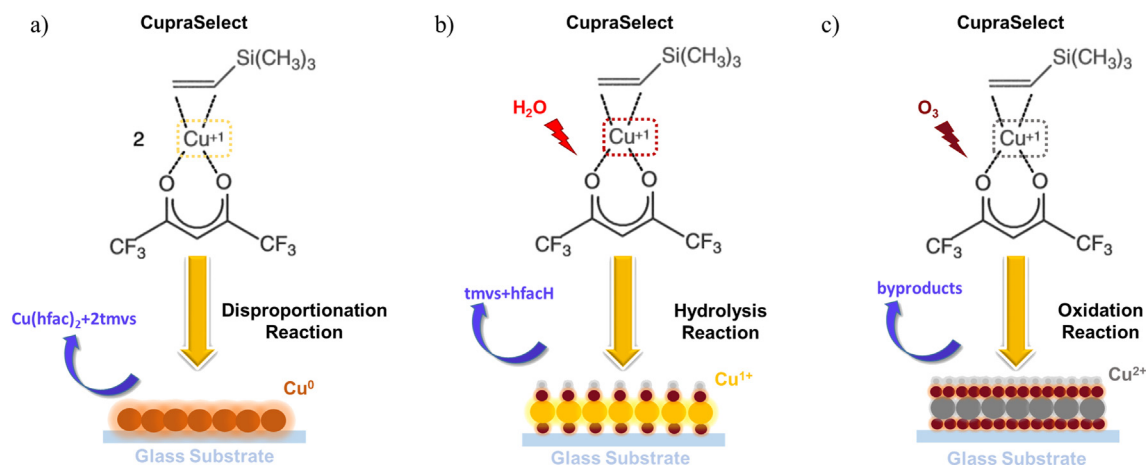


Fig. 1. Scheme of the reaction pathway for the deposition from $\text{Cu(I)}(\text{hftac})(\text{tmvs})$ precursor for the case of (a) metallic copper, (b) cuprous oxide, and (c) cupric oxide by varying the coreactant during the deposition process: N_2 , H_2O , and O_3 , respectively.

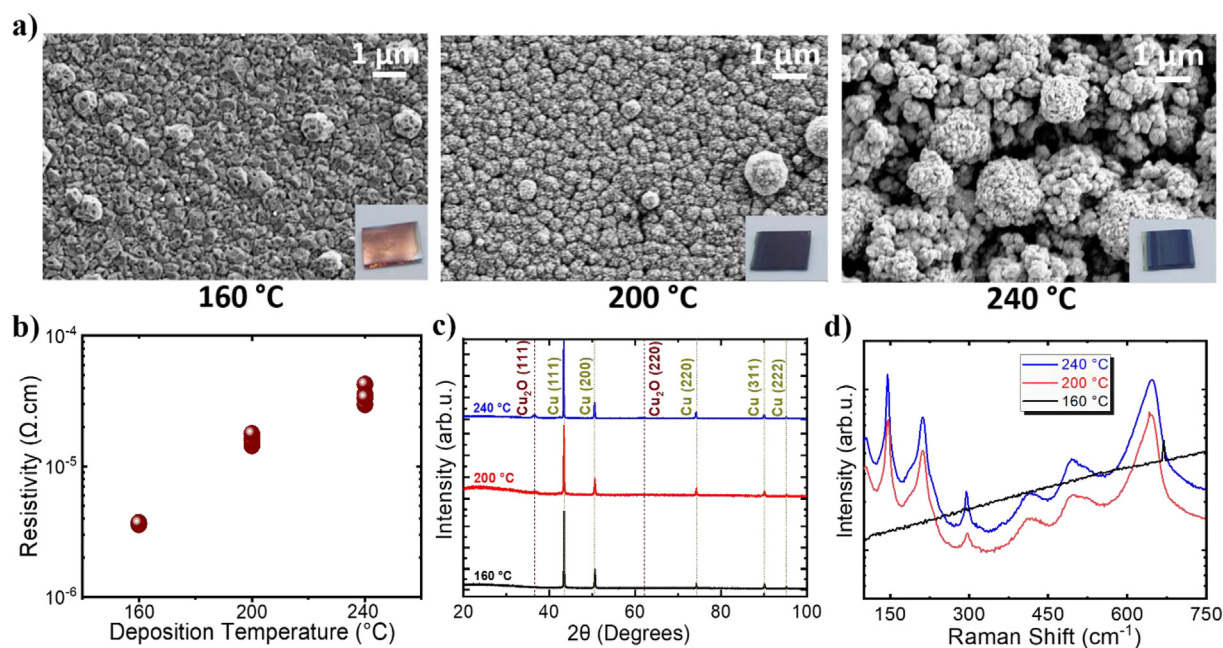


Fig. 2. (a) Top view scanning electron microscopy images, (b) four-probe resistivity measurements, (c) X-ray diffraction spectra, (d) Raman measurements, of copper films deposited on a glass substrate at 160 °C, 200 °C, and 240 °C for 8000 ALD cycles.

ablated film exhibits an important increase of the relative concentration of Cu^0 for samples deposited at 160 °C and 240 °C, respectively (Fig. 3d–f). The remaining percentage for samples deposited at low temperature (160 °C) is attributed mainly to $\text{Cu}(\text{OH})_2$ and not to the formation of other phases from Cu^{1+} and Cu^{2+} , while at higher deposition temperature (240 °C), there is a significant contribution in the percentage difference that is assigned to other copper phases. The presence of $\text{Cu}(\text{OH})_2$ in our films is explained by the fact that our deposition is performed in an open atmosphere, which can leave traces of water mainly on the surface. Indeed, the evolution of sample composition from surface to the bulk demonstrates the predominant contribution of the Cu^0 at the bulk level with still a small contribution of the $\text{Cu}(\text{OH})_2$. This presence is more predominant for samples deposited at 160 °C, as expected since hydroxides tend to dehydrate with time and/or temperature to yield the corresponding oxides [72,73].

TEM measurements have been carried out to study the microstructure and to further confirm the metallic nature of the copper layers prepared. The cross-section TEM image of a layer presenting a thickness of around 500 nm is shown in Fig. 4a. The grains grow in different orientations and show a dispersion in size. High-resolution TEM is shown in Fig. 4b, where the lattice spacing is 2.1 ± 0.3 Å, fitting the one of (111) metallic Cu. A selected-area diffraction pattern of the film is presented in Fig. 4c, where only diffraction rings corresponding to Cu can be observed.

3.2. Deposition of oxidized phases: Cu_2O and CuO

In our previous reports, we have shown that Cu_2O thin films can be obtained by combining $\text{Cu(I)}(\text{hftac})(\text{tmvs})$ with water. In the work of D. Muñoz-Rojas et al., high-quality, pure, and dense films were deposited at different temperatures without exceeding 225 °C. The

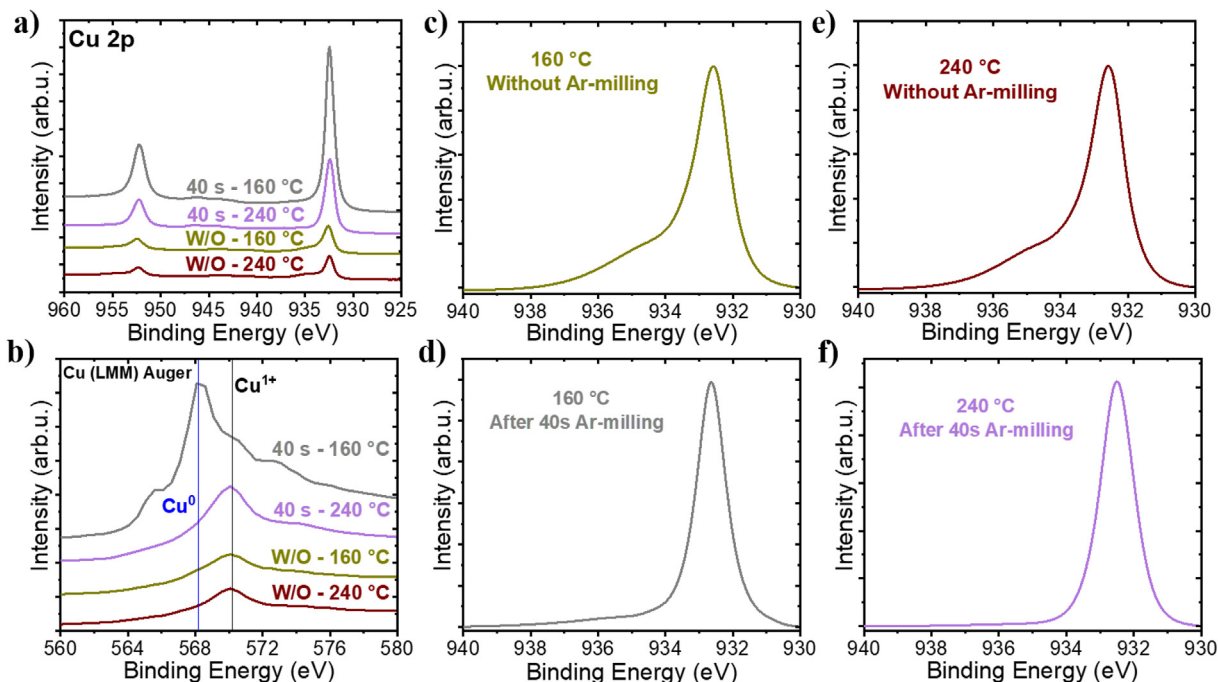


Fig. 3. X-ray photoelectron spectroscopy of Cu films (around 1 μm thick) deposited at 160 $^{\circ}\text{C}$ and 240 $^{\circ}\text{C}$ on glass substrates, (a) the Cu 2p, (b) Cu (LMM) Auger regions without (W/O) and with 40 s of argon milling, (c) a zoom view over the $\text{Cu}2p_{3/2}$ peaks without milling of samples deposited at 160 $^{\circ}\text{C}$ and (d) after 40 s of argon milling, and (e) at 240 $^{\circ}\text{C}$ without (W/O) milling and (f) after 40 s of argon milling.

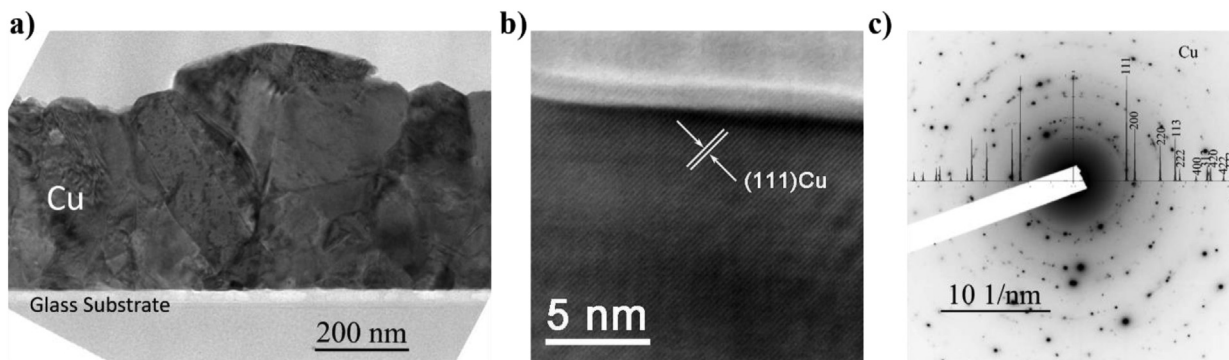


Fig. 4. (a) Cross-section transmission electron microscopy analysis of copper films deposited at 160 $^{\circ}\text{C}$ on a glass substrate, (b) high-resolution cross-section images of the copper film, and (c) electron diffraction of the copper metallic thin film.

deposition was achieved on glass and polymeric substrates [57]. In this section, we show that CuO thin films can be obtained by combining the same Cu precursor with ozone as coreactant (as opposed to just N_2 for Cu or H_2O for Cu_2O). Although Cu_2O can be easily oxidized to CuO by a thermal treatment at 300–1050 $^{\circ}\text{C}$ [74,75], our aim was to demonstrate that AP-SALD can be used to obtain pure as-deposited CuO thin films at low temperatures for applications that require low thermal budget processing without the need for a postannealing treatment.

The films have been deposited at 260 $^{\circ}\text{C}$ to ensure a sufficient thermal budget and prompt the oxidation reaction. Fig. 5a shows the SEM pictures of the as-deposited CuO film with an OR varying from 1500 ppm to 2500 ppm. A photograph of the films showing their color is presented at the bottom right part of the corresponding SEM image. As shown in the SEM images, the morphology of the films evolves with OR. At 1500 ppm, the film presents grains of around 200 nm with irregular shapes. Conversely, for 2000 and 2500 ppm, the grains size diminishes (around 50 nm for 2500 ppm)

and the films appear rougher. GI-XRD has been performed to investigate the film crystallinity and phase purity, as shown in Fig. 5b. The film obtained at a lower OR shows peaks from both CuO and Cu_2O . As expected, and inferred from the darkening of the films, the increase of the OR shows a drastic change, and only peaks from CuO can be observed. As shown in the figure, both crystallinity and texture are affected by the OR used, and therefore, a further optimization would be required depending on the properties or applications targeted.

In agreement with XRD data, Raman spectroscopy measurements reveal the presence of the vibrational modes of both Cu_2O and CuO for 1500 ppm. The CuO Raman modes appearing in the following wavelengths: 282 cm^{-1} , 330 cm^{-1} , and 616 cm^{-1} (Fig. 5c). For higher OR only, the Raman modes of CuO were observed.

In order to further probe the phase purity of the films, XPS measurements were carried out on samples deposited with OR of 1500 ppm and 2500 ppm. Fig. 6a shows the XPS pattern in the Cu 2p regions for both films. The results show that both phases are

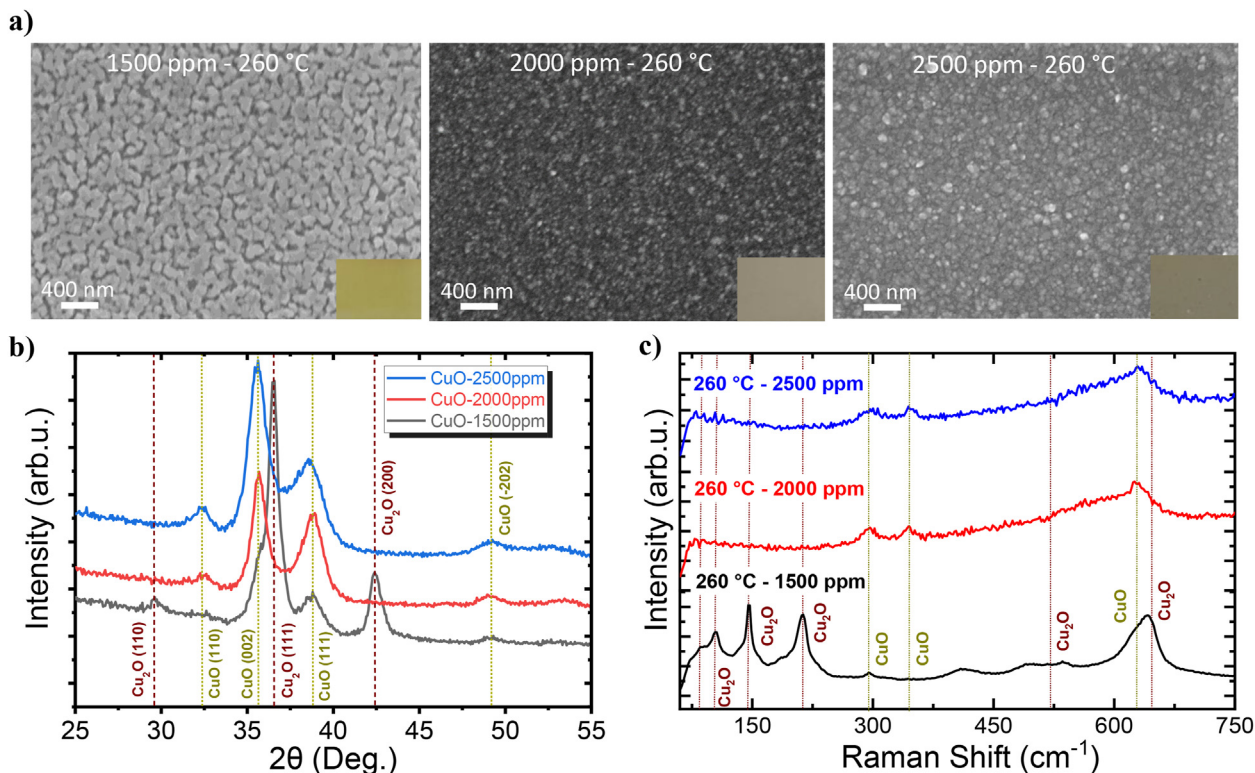


Fig. 5. (a) scanning electron microscopy images, (b) GI-XRD patterns, and (c) Raman spectra, of cupric oxide (CuO) films deposited on a glass substrate at 260 °C with 1500 ppm, 2000 ppm, and 2500 ppm for 1000 cycles, which is respectively equivalent to 170 nm, 160 nm, and 160 nm. (CuO peak contributions are highlighted in dark green and Cu₂O ones in dark red).

present for 1500 ppm since Cu¹⁺ and Cu²⁺ satellites are visible. Conversely, a clear contribution of only the Cu²⁺ satellites is obtained for the film deposited with an OR of 2500 ppm. The peak decomposition, in Fig. 6b–c, shows the contribution of each species. For the low OR film, a small presence of the Cu(OH)₂ is observed. Then, similar percentages of Cu¹⁺ and Cu²⁺ are obtained. Further, the film deposited with higher OR shows the presence of mainly Cu²⁺.

Although the XRD, TEM, Raman, and XPS results all seem to indicate that the different phases are obtained pure for optimized deposition parameters, XAS studies were as well performed to probe the bulk of the films with a higher sensitivity. The X-ray absorption near edge spectra obtained for Cu, Cu₂O, and CuO thin films deposited with optimized parameters (at 160 °C, 220 °C, and 260 °C with 2500 ppm OR) are presented in Fig. 7a. The spectra of

the three phases are clearly different from one another. As reported previously, the energy at which the edge absorption is observed increases for increasing Cu oxidation state [76], as is the case in the obtained spectra for our Cu, Cu₂O and CuO thin films. In addition to the edge energy, the shape of the peak can be correlated with the oxidation state of the Cu atoms and thus the phase present. The spectra obtained clearly show a decrease of the pre-edge feature as the oxidation state increases from Cu to CuO. In addition, the white line of the copper film is sharper, and well-defined oscillations are observed at higher energies related to the pure metallic copper phase [77]. Treated extended X-ray absorption fine structure data are presented in Fig. 7b. The Fourier transform of the spectra provide the average radial distribution function around the copper atoms (i.e., mean distance between Cu atoms and the different coordination spheres; Cu, O, Cu, etc.). Two main shells are identified

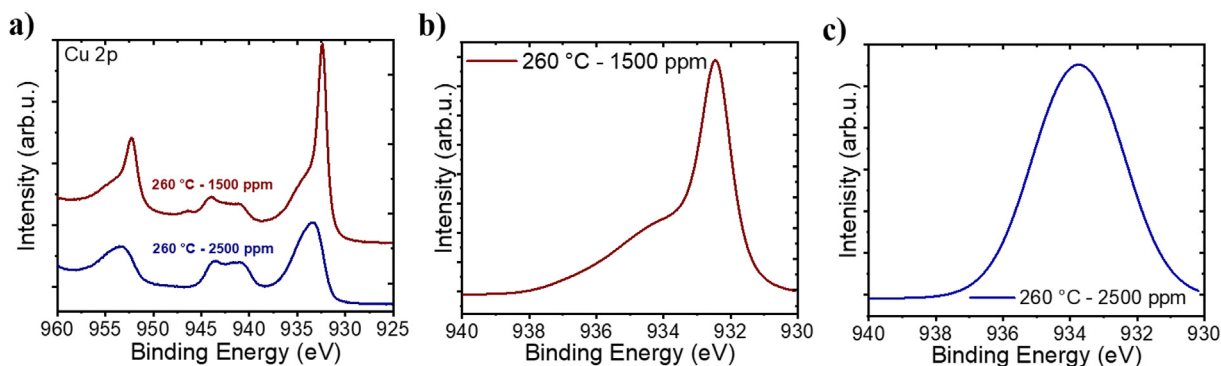


Fig. 6. X-ray photoelectron spectroscopy of CuO films deposited at 260 °C on a glass substrate with 1500 ppm and 2500 ppm, (a) the Cu 2p, and fitting of the Cu2p_{3/2} peaks for samples deposited with (b)1500 ppm, and (c) 2500 ppm. The orange numbers represent the amount the Cu⁰, Cu¹⁺, Cu²⁺, Cu(OH)₂ within the films.

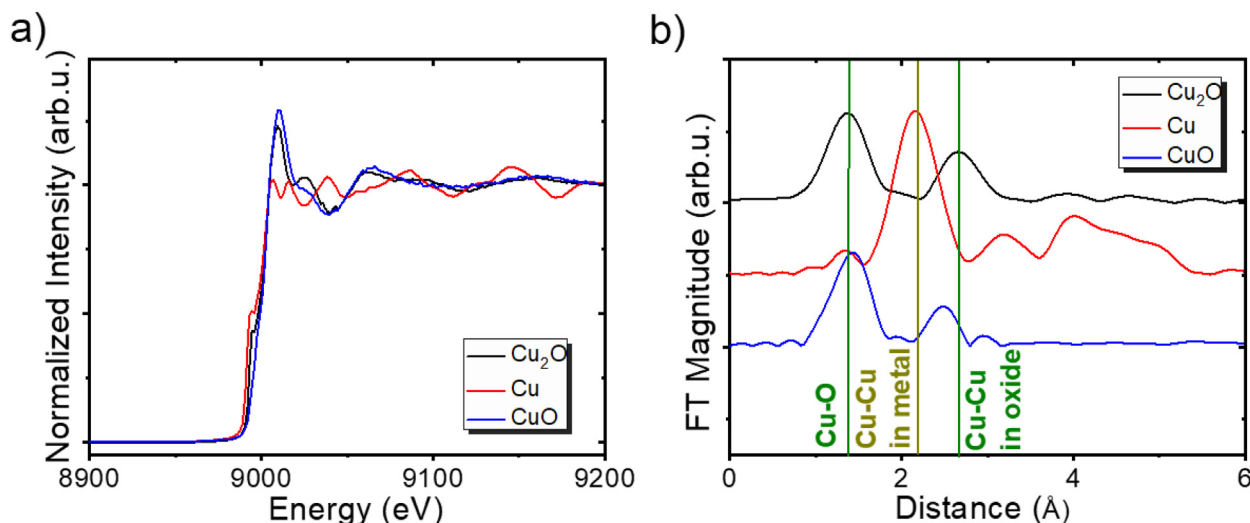


Fig. 7. Cu K-edge X-ray absorption spectroscopy data for the Cu_2O , CuO , and Cu thin films. (a) X-ray absorption near edge region, (b) amplitude of the Fourier transform (FT) of the extended X-ray absorption fine structure (EXAFS) data (selected subset of data).

Table 1
Summary of the main properties of pristine films.

Grown films	Film color	Deposition temperature ($^{\circ}\text{C}$)	Coreactant	Crystallinity	Electrical resistivity ($\Omega\cdot\text{cm}$)
Copper	Brown	160	Nitrogen	Polycrystalline: Cu (111)	3.7×10^{-6}
	Black/blue	200	Nitrogen	Polycrystalline: Cu (111) & Cu_2O (111)	1.5×10^{-5}
	Black/blue	240	Nitrogen	Polycrystalline: Cu (111) & Cu_2O (111)	3.6×10^{-5}
CuO	Green	260	Ozone – 1500 ppm	Polycrystalline: CuO (002) & Cu_2O (111)	0.19
	Gray	260	Ozone – 2000 ppm	Polycrystalline: CuO (002)	0.2
	Dark gray	260	Ozone – 2500 ppm	Polycrystalline: CuO (002)	1.91
Cu_2O [57,78]	Yellow	125–225	Water	Polycrystalline Cu_2O (111)	0.4–1000

for Cu_2O and CuO , a first one related to the oxygen first neighbors ($\text{Cu}-\text{O}$) and a second shell related to the closest copper neighbors ($\text{Cu}-\text{Cu}$). But in the case of CuO , a splitting of the second shell to two contributions (related to the third order O neighbors) together with a clear shift to lower distance of the ($\text{Cu}-\text{Cu}$) neighbor confirm the phase purity. In the case of metallic copper, a clear shell distance related to the ($\text{Cu}-\text{Cu}$) neighbor in metal is observed confirming the phase purity of the metallic copper [63]. XAS data thus confirm the phase purity of the different thin films. Table 1 Summarizes the relevant data obtained for the three obtained phases highlighting the low values of resistivity of copper metal quite close to the values of the bulk. Such low resistivity values are noteworthy, especially considering that an atmospheric and low-temperature chemical deposition method is used. These results show the potential of using these materials for the low-cost fabrication of flexible electronic and optoelectronic devices, paving the way for new developments for cost-effective applications.

4. Conclusion

In summary, we have demonstrated the possibility to obtain the three main phases of copper (Cu , Cu_2O , and CuO) at low temperature ($160\text{ }^{\circ}\text{C}$ – $260\text{ }^{\circ}\text{C}$) and using the same metallic precursor with our open-air AP-SALD system. By only changing the nature of the coreactant (O_3 , H_2O , or N_2) and adjusting the deposition temperature, the three phases were obtained with a high degree of purity, as shown by XPS and XAS. Remarkably, Cu thin films are obtained without the need for a reducing agent, and for the first time, CuO has been grown using AP-SALD process. The formation of Cu , Cu_2O ,

and CuO thin films occurs through disproportionation, hydrolysis, and oxidation reactions, respectively. Metallic copper presents a low resistivity of $\sim 3.7 \times 10^{-6} \Omega\cdot\text{cm}$, while resistivity values of $100 \Omega\cdot\text{cm}$ and $0.2 \Omega\cdot\text{cm}$ have been measured for the Cu_2O and CuO thin films. These results open prospects for the low-cost manufacturing of flexible electronic and optoelectronic devices paving the way for further development related to future integrations into devices.

Author contributions

D.M.R. and D.B. obtained the funding. D.M.R. conceived the experiments. A.S. performed experiments and most characterizations. M.W. performed XPS characterization, J.L.S. and J.R.Z. performed XAS characterization and analysis. Experimental data was analyzed under the supervision of A.K.-C., G.C., D.B., and D.M.-R. All authors contributed to the interpretation of the results. A.S. and D.M.R. wrote the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtchem.2023.101431>.

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